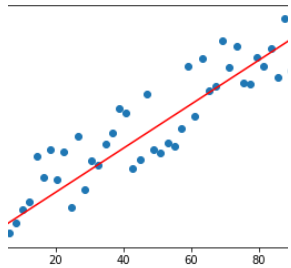


A4.3.1 Linear regression



A4.3.1 Explain how linear regression is used to predict continuous outcomes

4.3.1 explains linear regression, a Higher Level (HL) machine learning approach, used to predict continuous outcomes. This technique focuses on the relationship between independent (predictor) and dependent (response) variables. It highlights the significance of the slope and intercept within the regression equation. The effectiveness of how well the model fits the data is commonly assessed using measures like r^2 .

Introduction

So far, we've prepared our data and explored different learning paradigms. Now, we'll build a model that can predict a numerical outcome. **Linear Regression** is a fundamental and powerful tool for understanding and predicting relationships in data.

What is Linear Regression?

Linear Regression is a supervised learning algorithm used to predict a **continuous outcome** (a number) based on one or more predictor variables. Its goal is to find the single straight line that best represents the relationship between the variables.

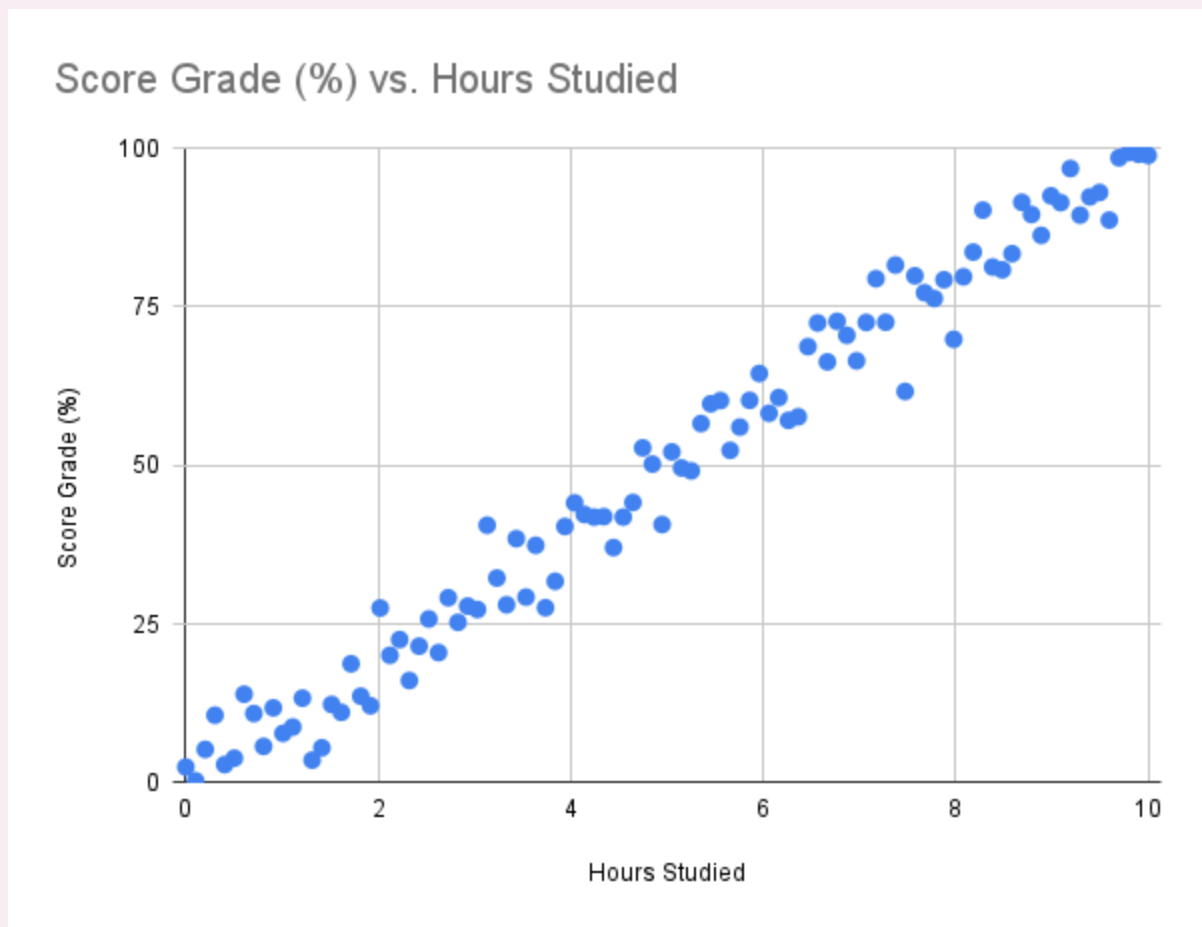
- **Continuous Outcome:** A value that can be any number within a range, like temperature, a person's height, or the price of a house. This is different from classification, where we predict a category (e.g., 'Cat' or 'Dog').

The Relationship Between Variables

Linear regression works by modeling the relationship between two types of variables:

1. **Independent Variable (or Predictor):** This is the variable we use to make a prediction. It's the "cause" or the input. On a graph, this is always the **x-axis**.
 - *Example:* Hours a student studies.
2. **Dependent Variable (or Response):** This is the variable we are trying to predict. It's the "effect" or the outcome. On a graph, this is always the **y-axis**.
 - *Example:* The final test score a student gets.

The core idea is that the **dependent variable depends on the independent variable**. The test score depends on the hours studied.



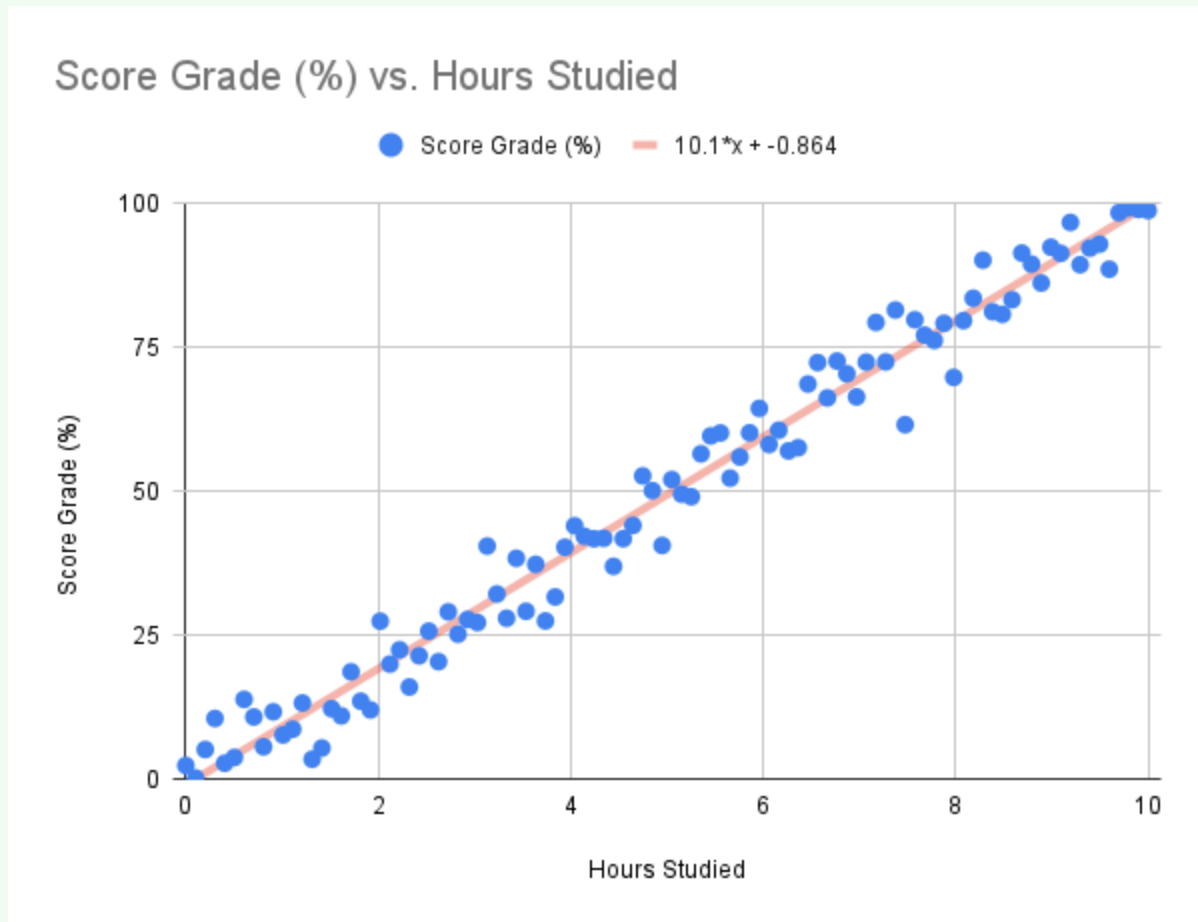
The Significance of the Slope and Intercept

The goal of linear regression is to draw a "**line of best fit**" through the data points on a scatter plot. This line is represented by the familiar equation:

$$y = mx + c$$

In machine learning, each part of this equation has a very important real-world meaning:

- **c (The Y-Intercept):** This is the predicted value of y when x is 0. It's the baseline or starting point of our model.
 - *In our example:* The y-intercept is the predicted test score for a student who studied for **0 hours**.
- **m (The Slope):** This tells us how much the y variable is expected to change for every one-unit increase in the x variable.
 - *In our example:* If the slope is around 10, it means that for every **one additional hour** a student studies, we predict their test score will **increase by 10 points** on average.



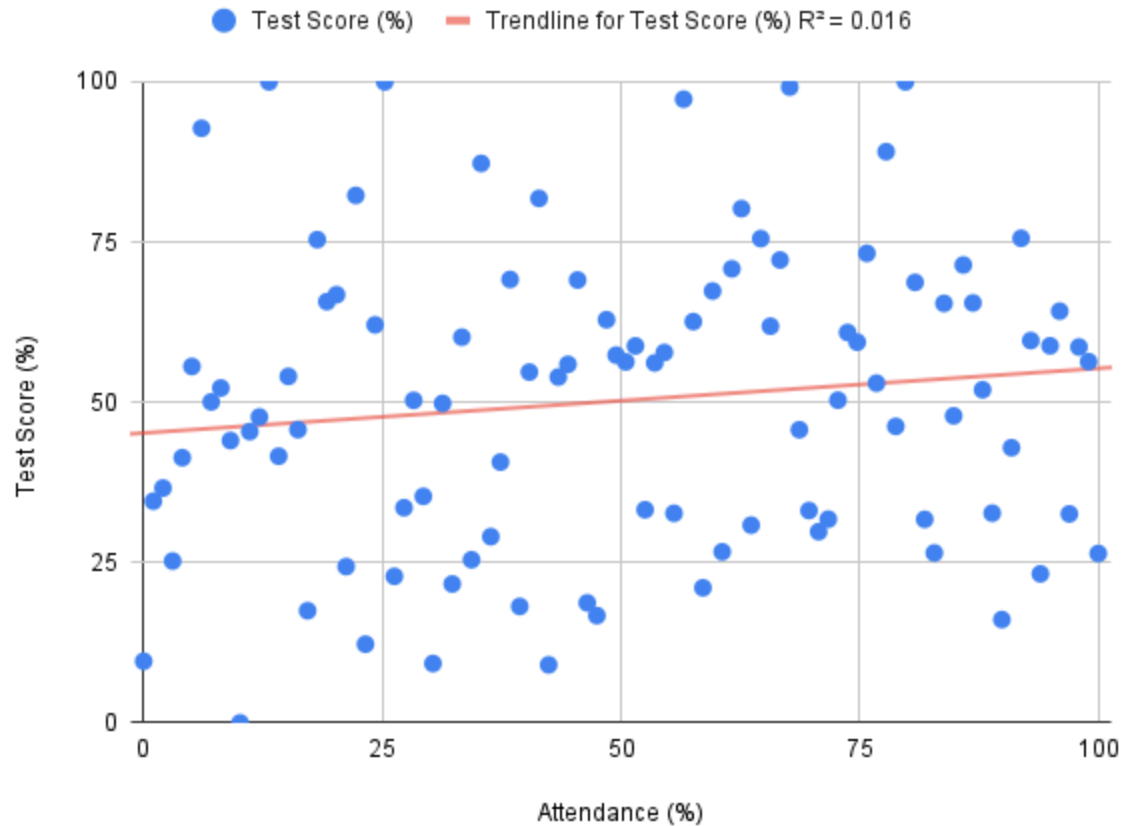
How Well Does the Model Fit the Data?

Just because we can draw a line doesn't mean it's a good one. We need a way to measure how well our line actually represents the data. The most common measure for this is **R-squared (r^2)**.

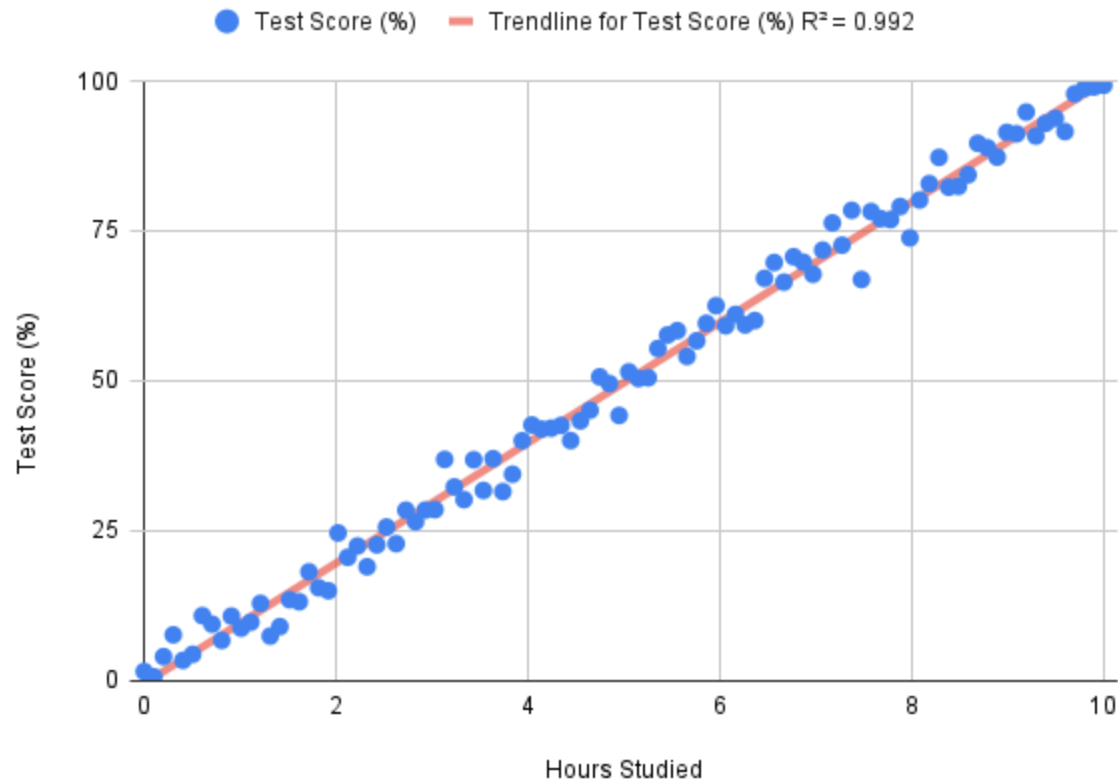
- **What it is:** R-squared is a statistical measure between 0 and 1. It tells us the **proportion of the variance** in the dependent variable that is predictable from the independent variable.
- **In simple terms:** R-squared is a percentage score that tells you how much of the "story" in the data your line explains.
 - An **r^2 of 0** means your model explains none of the variability. It's useless.

- An r^2 of 1 means your model explains all of the variability. It's a perfect fit.
- An r^2 of 0.85 means that 85% of the changes in the test scores can be explained by the hours studied, according to our model. This is generally considered a very good fit.

Test Score (%) vs. Attendance (%)



Test Score (%) vs. Hours Studied



Flashcards

Use this Quizlet to learn all the key terms in this section.

There is no set with the ID you specified

 Try a search instead

Or browse subjects

Science



Languages



Arts and
Humanities



Math



Social Science



Other



Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

Linear Regression is a supervised learning technique used to predict what kind of outcome?

A. ☐ A discrete category

B. ☒ A continuous value

C. ☐ An unsupervised cluster

D. ☐ A class label

2

In a model predicting a car's price based on its mileage, "mileage" is the:

A. ☐ R-squared value

B. ☐ Y-Intercept

C. ☒ Independent Variable

D. ☐ Dependent Variable

3

The equation for a model is $\text{Score} = 4 * \text{Hours} + 50$. What is the predicted score for a student who studies for 0 hours?

A. ☐ 54

B. ☐ 46

C. ☒ 50

D. ☐ 4

4

In the model $\text{Price} = -100 * \text{Age} + 5000$, what does the slope of -100 represent?

A. ☐ The starting price is -\$100.

- B. ☐ The price of a new item is \$100.
- C. ☐ For every year older, the price increases by \$100.
- D. ☒ For every year older, the price decreases by \$100 on average.

5

A data scientist reports that their linear regression model has an r^2 value of 0.15. This indicates that:

- A. ☐ The slope of the line is 0.15.
- B. ☒ The model explains very little of the variability in the data.
- C. ☐ 15% of the predictions are correct.
- D. ☐ The model is a very good fit for the data.

6

The "line of best fit" is the line that:

- A. ☐ Passes through the most data points possible.
- B. ☐ Has the steepest possible slope.
- C. ☒ Minimises the overall distance from the line to all data points.
- D. ☐ Connects the first and last data points.

7

Which of the following problems is a suitable task for Linear Regression?

- A. ☐ Classifying an email as 'spam' or 'not spam'.

- B. ☐ Identifying a person's face in a photo.
- C. ☐ Grouping customers into different market segments.

D. ☒ Predicting the temperature tomorrow based on today's temperature.

8

The variable that is being predicted in a linear regression model is known as the:

- A. ☐ Predictor Variable
- B. ☐ Slope
- C. ☐ Independent Variable

D. ☒ Dependent Variable

9

A perfect linear regression model that explains 100% of the variation in the data would have an r^2 value of:

A. ☐ 100

B. ☒ 1

C. ☐ It depends on the slope.

D. ☐ 0

10

In the equation $y = mx + c$, the 'c' term represents the:

A. ☐ R-squared value

B. ☐ Y-Intercept

C. ☐ Independent Variable

D. ☐ Slope

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